Pelka 2021 colorectal cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

64 tumours integrate to one continuous manifold (same-sample neighbour fraction 0.71→0.10, robust across θ 2–10); a single connected component after island removal.

Gate 2 — module geography → GO (non-branching)

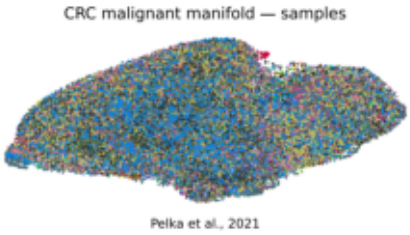
Non-branching (2 tips, 0 forks). Shared developmental gradient with neural crest as the inverse terminal end: pSom↔Crest spatial r=−0.89; along pseudotime pSom ρ=−0.79, Crest ρ=+0.83.

Gate 3 — cross-repression → NA

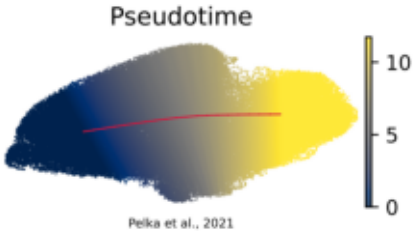
No bifurcation → no hub to test. Cross-repression gate not applicable to the non-branching case.

STEP FIGURES

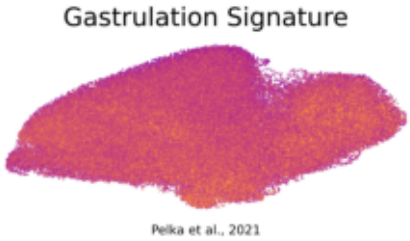
Manifold — samples



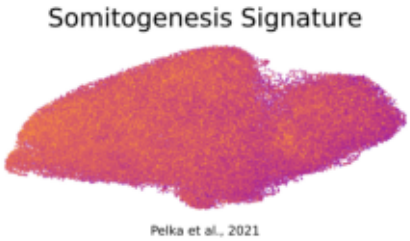
Pseudotime



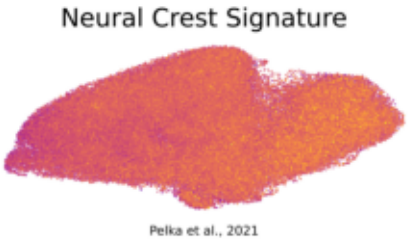
Gastrulation



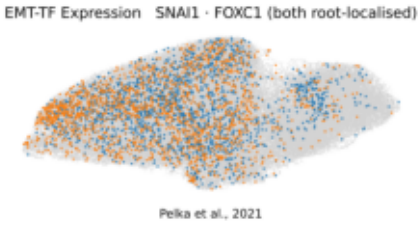
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Colorectal cancer (Pelka et al., 2021), 10x; 108,999 published-malignant cells across 64 tumours, all samples above the atlas 100-cell floor; 108,979 after removing 20 detached-island cells.

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest developmental axis. Gate 1 GO / Gate 2 GO / Gate 3 NA. No competing arms, so no hub r. EMT-TFs SNAI1 and FOXC1 both localise to the root; canonical ZEB1/TWIST1 below the 3% floor.

Parameter choices

Min cells/sample	≥100 floor (gap-rule degenerate 1.58)	**Integration**	Harmony θ=10; n_pcs=20 (4,556 HVGs)
Scope	whole manifold (proliferation orthogonal)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100 (SA 2.99)
Root node	somitogenesis/gastrulation-high tip	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	SNAI1 · FOXC1 (both root-loaded)		